

DATA ANALYTICS LEVEL III

Day 7: Filtering Like an Analyst — Smart Filters, Sorting & DISTINCT

Student Activities

How to Use This Document

You are the data analyst at a Philippine electronics retailer. Today's activities are written the way real requests arrive — in plain English, from your manager. For EVERY question: (1) write the SQL you ran, (2) write the answer (row count, names, or values as asked). Connect to Day07_practice.db first (new SQLTools connection named Day07).

Activity	What You Submit	Time
Activity 1: Filter Drills	Queries + answers for all 10 drills	70 min
Activity 2: The Manager's Requests	Queries + answers for all 9 requests	70 min
Learning Journal Entry #6	Your journal reflection	20 min

Activity 1 — Filter Drills

Goal: Master AND/OR, IN, BETWEEN, and LIKE one condition at a time.

Before you run each query, PREDICT how many rows it will return, write your guess, then run it and compare. (Guessing first is how you build analyst instinct.)

Part A — AND / OR / Parentheses

- Sales department employees earning MORE than ₱30,000. (Show first_name, salary.) Predicted: ____ Actual: ____
- Employees based in Cebu OR Davao. (Show first_name, city.) Predicted: ____ Actual: ____
- Accessories priced UNDER ₱1,000. (Show product_name, price.) Predicted: ____ Actual: ____
- Employees in Sales OR Marketing who earn at least ₱35,000. (Show first_name, department, salary.) Careful — you need parentheses! Predicted: ____ Actual: ____

Part B — IN and BETWEEN

5. Rewrite drill 2's idea for the capital: employees in Manila or Makati — but use IN, and just COUNT them. (Hint: SELECT COUNT(*) ...)
6. Employees earning BETWEEN ₱30,000 and ₱40,000. (Show first_name, salary.) Before running: will Hector (30,000) and Ramon (40,000) appear? Write your prediction, then check.
7. Products with stock BETWEEN 10 and 20 units. (Show product_name, stock_qty.)

Part C — LIKE (patterns)

8. Employees whose first_name ENDS with the letter “a”. (Show first_name.)
9. Products with “USB” ANYWHERE in the name. (Show product_name.)
10. Products whose name STARTS with “Laptop”. (Show product_name.) Look closely at the result — is every row actually a laptop? Write one sentence about what you notice.

Submit as: Day07_Activity1_[YourName].docx (or .sql with comments)

Activity 2 — The Manager's Requests

Goal: Translate real plain-English requests into SQL using ORDER BY, LIMIT, DISTINCT — and everything from Activity 1.

Each item below is a message from your manager. Answer with a query and the result. Watch for English words like “and” that do NOT mean SQL AND!

11. “What cities do we actually have staff in? Just the list of cities, no repeats.” (Bonus: do the same for product suppliers — how many unique suppliers do we buy from?)
12. “Send me all employees sorted by salary, highest first.” (Show first_name, last_name, salary.) Who is at the top?
13. “Actually, just the TOP 3 earners.” (Same columns, one small change.)
14. “What are our 3 CHEAPEST products?” (Show product_name, price.)
15. “Give me the product list grouped by category, and inside each category show the most expensive first.” (Show product_name, category, price — this is a two-column sort.)
16. “Which Electronics cost ₱10,000 or more? Most expensive first.” (Show product_name, price.)
17. “Re-stock report please: everything with fewer than 15 units left, lowest stock at the top, and include the supplier so purchasing knows who to call.” (Show product_name, stock_qty, supplier.)
18. “HR is asking: who did we hire in 2022?” (Show first_name, hire_date. Hint: hire_date is text in YYYY-MM-DD format — a LIKE pattern works.)
19. CHALLENGE — “Send me the Manila and Makati staff earning above ₱30,000, highest paid first.” (Show first_name, city, salary.) This one needs IN + AND + ORDER BY in a single query. How many rows, and who leads the list?

Submit as: Day07_Activity2_[YourName].docx (or .sql with comments)

Learning Journal Entry #6

Goal: Reflect on thinking in conditions and patterns.

1. A manager says “show me the staff in Cebu and Davao.” Why would WHERE city = 'Cebu' AND city = 'Davao' return zero rows? What should the query use instead?

2. What is the difference between LIKE 'USB%' and LIKE '%USB%'?

3. BETWEEN 30000 AND 40000 — are 30,000 and 40,000 included or excluded? How did you verify it today?

4. My confidence with SQL filtering today (1 = lost, 5 = very confident), and why:

Submit as: Day07_Journal6_[YourName].docx

Submission Checklist

- [] **Activity 1:** all 10 drills with predictions, queries, and answers
- [] **Activity 2:** all 9 manager requests with queries and answers
- [] Learning Journal Entry #6